

Date: 01/19/16  
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**Weekly Report 2 due January 19, 2016**

This week I spent most of my time researching parts to buy and trying to determine if my initial design (a robot that catches a thrown ball) is feasible given the time frame and relative budget range. Early on in the week, I finished the robot abstract assignment, coming up with a very preliminary bill of materials and list of intended functionality. As the weekend approach, I was also successfully able to run OpenCV programs on the Jetson TK1 development board – however I was a little disappointed in the frame rate that I was receiving. From what I could gather online, the board was not utilizing all of its resources (GPU, MIPI input, etc.), so I plan to learn how to exploit these features this week.

It became obvious that the intended functionality was quite ambitious, so to ensure that the idea was at all feasible, I began work to create a computer simulation of the robot in the Unity3d game engine (which has built-in physics simulation). I will use the simulation primarily to invent an algorithm in OpenCV which I know will allow the robot to catch the ball; from there, I will program in a simulated frame rate and delay in motor communication and starting/stopping time in order to determine the lower bound on hardware needed for the design to work. While I did not finish the simulation, my goal is to do so early this next week and determine if the current project is feasible or if I will be required to pursue an alternate design.

As the simulation will prove crucial to the specification of parts needed, I decided to delay ordering any actuators/sensors until I was sure what I needed. If all goes well, I will be able to order the parts late this following week with relative certainty that they will be adequate to realize the final design.